

Philipp Schröppel

PHD STUDENT IN COMPUTER VISION AT THE UNIVERSITY OF FREIBURG

Freiburg i. Br., Germany

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Personal Information

Name Philipp Schröppel
Address 79106 Freiburg i. Br., Germany
Mail schroepp@cs.uni-freiburg.de
Birthyear 1992
Nationality German

Research

My broad research area is **3D reconstruction in terms of 3D geometry, ego-motion and object motion**. A particular focus is robust application on arbitrary real-world data. To this end, I am interested in:

- multi-view depth estimation,
- single-view depth estimation,
- optical flow estimation,
- scene flow estimation,
- depth-from-video,
- implicit 3D scene representations (neural fields).

Recently, I worked on **3D generation using diffusion models**. Currently, I am working on **3D object-centric learning**, with the long-term goal of making 3D generative models more effective.

Publications

CONFERENCE PUBLICATIONS (* denotes equal contribution)

- ECCV 2024 Silvio Galesso*, **Philipp Schröppel***, Hssan Driss, Thomas Brox. *Diffusion for Out-of-Distribution Detection on Road Scenes and Beyond*. European Conference on Computer Vision, 2024.
- CVPR 2024 **Philipp Schröppel**, Christopher Wewer, Jan Eric Lenssen, Eddy Ilg, Thomas Brox. *Neural Point Cloud Diffusion for Disentangled 3D Shape and Appearance Generation*. Conference on Computer Vision and Pattern Recognition, 2024.
- 3DV 2022 **Philipp Schröppel**, Jan Bechtold, Artemij Amiranashvili, Thomas Brox. *A Benchmark and a Baseline for Robust Multi-view Depth Estimation*. International Conference on 3D Vision, 2022.
- GCPR 2022 Leonhard Sommer, **Philipp Schröppel**, Thomas Brox. *SF2SE3: Clustering Scene Flow into SE(3)-Motions via Proposal and Selection*. German Conference on Pattern Recognition, 2022.

WORKSHOP PAPERS (* denotes equal contribution)

- CVPR 2021 Julia Guerrero-Viu*, Sergio Izquierdo*, **Philipp Schröppel**, Thomas Brox. *Semi-Supervised Disparity Estimation with Deep Feature Reconstruction*. Women in Computer Vision Workshop, 2021.

Education

PhD Student Computer Vision and Deep Learning

COMPUTER VISION GROUP FREIBURG, HEADED BY PROF. THOMAS BROX

Working on 3D reconstruction and 3D generation with a focus on robust application to real-world data. (Python, PyTorch, Tensorflow)

Freiburg i. Br.
since 01/2019

Information Systems Engineering, Grade: 1.7

TU DRESDEN

- Degree: Dipl.-Ing. for Information Systems Engineering (equivalent to M.Sc.).
- Thesis, SICK AG, Waldkirch, Grade: 1.6.
Title: *Detection and Mapping of Obstacles for an Automated Guided Vehicle Using a 3D Sensor*. (C++)
- Student research thesis (equivalent to Bachelor thesis), Chair for Automation Engineering, Grade: 1.0.
Title: *Developing a SLAM Algorithm for an Omnidirectional Robot Using a ToF Depth Sensor*. (C++, ROS)

Dresden
10/2012 – 07/2018

Electrical Engineering

TU DRESDEN

Switched to information systems engineering after one year in order to enroll for more computer science courses.

Dresden
10/2011 – 10/2012

Abitur, Grade: 1.5

MARKGRAF-GEORG-FRIEDRICH-GYMNASIUM

Kulmbach
2003 – 2011

Work Experience

Research Internship

GEOMETRIC REPRESENTATION LEARNING GROUP WITHIN THE MAX PLANCK INSTITUTE FOR INFORMATICS
3D object-centric learning. (PyTorch)

Saarbrücken
12/2024 – 05/2025

Internship

SICK AG
Gesture recognition using data of a 3D time-of-flight sensor. (C++)

Waldkirch (near Freiburg i. Br.)
5/2017 – 11/2017

Research Assistant

CHAIR OF AGRICULTURAL SYSTEMS TECHNOLOGY, TU DRESDEN

- Linux server administration.
- Extending a system for the management of agricultural processes and the analysis of acquired sensor data. (PHP, JavaScript)

Dresden
7/2016 – 2/2017

Research Assistant

CHAIR OF PROCESS CONTROL SYSTEMS, TU DRESDEN

Development of Android apps building on semantic web technologies. (Java)

Dresden
11/2013 – 9/2014 and 4/2015 – 7/2015

Teaching & Mentoring

2020	Master Thesis: Lal Jose. <i>Monocular Camera Tracking for Driving Scenarios</i> .
2020 - 2021	Master Project: Saurav Shanu. <i>Stixel Prediction with an End-to-end deep Network</i> .
2020 - 2021	Master Project: Sergio Izquierdo Barranco. <i>Deep Feature Reconstruction for Disparity Estimation with DispNet</i> .
2020 - 2021	Master Project: Julia Guerrero-Viu. <i>Semi-Supervised Domain Adaptation for Disparity Estimation with DispNet</i> .
2021	Teaching Assistant: Image Processing.
2021	Master Thesis: Julia Guerrero-Viu. <i>Improving Deep Feature Representations for Self-Supervised Training of Disparity Estimation</i> .
2021	Teaching Assistant: Statistical Pattern Recognition.
2021 - 2022	Master Thesis: Saurav Shanu. <i>Monocular 3D Object Detection and Bird's Eye View Generation in Driving Scenarios</i> .
2021 - 2022	Master Thesis: Leonhard Sommer. <i>From Pixel Matching to Dynamic Rigid Objects</i> .
2023	Master Thesis: Saiprasad Barke. <i>Analysing Multi-view Depth Estimation in a Common Framework</i> .
2023	Master Thesis: Tom Wellinger. <i>Bootstrapping Single-view Depth Estimation via Multi-view Depth Estimation</i> .
2023 - 2024	Master Project: Achim Wimme. <i>Depth-from-video Estimation with a RAFT Model Architecture</i> .
2023 - 2024	Teaching Assistant: Computer Vision.
2024 - 2025	Master Thesis: Achim Wimme. <i>A Systematic Comparison of Different Deep Learning Pose Estimation Strategies</i> .

Skills

Programming	Python, PyTorch, Numpy, Tensorflow, C++, C, Java, Haskell, JavaScript, HTML, CSS, PHP, OpenCV, ROS, Eigen
Knowledge	Computer Vision, Deep Learning, Machine Learning, Robotics (e.g. SLAM, 3D Mapping), 3D sensors, Software engineering
Miscellaneous	Linux administration, Git, $\LaTeX$, Android app development
Languages	German (<i>Native language</i>), English (<i>C1</i>), French (<i>B1 - B2</i>)

Activities

Sports	Preferably, I spend my free time outdoors and in company of my friends. We often go climbing, as much as possible outdoors, but also indoors in a climbing gym. Apart from climbing, I like running and cycling and go ski touring whenever I have the possibility to go to the Alps.
Travelling	All of this works well with my favour for travelling. Doing so, I like to live and move by simple means, for example with bike and tent.
Friends	On more quiet days, I also like to stay at home and read a book, or to meet friends and do something together that does not necessarily have to involve anything sports-related.